

ActInf GuestStream 066.2 ~ Adrian Bejan, "Freedom and Evolution"

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hello and welcome this is active inference guest stream 66.2 on February 15 2024 we're very much looking forward to this followup on a very delightful 66.1 so thank you Susan Hasty for organizing and helping with this to you for the introduction oh unmute and continue thank you okay thank you Daniel um and thank you Adrien for coming back um last uh our last session we covered design in nature your book on design in nature and um and this is my favorite book of all which is what we're going to cover today um which is evolution and freedom so I'm going to turn it over to you and I look forward to this presentation and our discussion afterwards okay thank you for the invitation uh especially uh in view of the fact that this is my second um um uh appearance on your um um show um the um the written version of uh of our discussion is in this book which is still relatively fresh and um I prepared a few slides for the next 20 or 25 minutes uh the first slides are A Brief Review of my previous um presentation which is about the fact that uh the phenomenon that um screams to be recognized uh in nature or for that matter in physics is the um uh appearance and then the evolution of the flow configuration the flow configuration as you can see here unites the animate with the inanimate in other words it does away with the dichotomy inherited from the church and with that idea going forward everything that flows and moves on Earth is driven by um by a power which comes from the consumption of fuel and food um and there flow architectures most of them natural that function as engines for example the the atmosphere the hydrosphere all these things that move are driven by power and as they as they move and as they morph they dissipate the power and I call the dissipated features brakes and on the right hand side you see a summary of everything uh which means the Geo the bio and the socio are of morphing meaning evolving flow architectures the um the statement that emerges after all these observations again it's based on observations in other words this law of evolution in nature is of empirical origin like all of laws of physics and um and um it says that it actually defines the life state which is our subject today for for a flow system to persist in time it must evolve with

freedom such that it provides greater access so the phenomenon is the evolution of flowing form uh which is actually the time Arrow or the definition of future as something different than the present and the past the keywords are only two the ones that will um give us support in the conversation so uh the the definition of Evolution from the dictionary is changes that occur in a discernable Direction in time um that direction is known by many names objective purpose go there are many other such words that uh that represent Common Sense Common Sense Freedom uh are physical features that allow an entity we call that system and theonomics to exhibit changes um Freedom can be measured in terms of the number of degrees of freedom for example uh the Piston moving in and out of the cylinder of the engine has one degree of Freedom which is the position of the Piston um other uh configurations the natural ones have innumerable degrees of freedom you can look at the direction in which your cat is uh is moving and um and anyway the freedom description is a very very thick book of examples uh the pun sign is that without this physical feature called Freedom uh there can be no Evolution and no design and for that matter no no direction in time uh the book has 11 chapters uh time is short so I will if I'm lucky I'll I'll touch on the the the chapters lighted in red and I'll take take them one at a time economies of scale is an observation from economics uh coming from um uh design in nature this observation is predictable uh in the upper left hand corner uh this is a classroom exercise you can show easily that if a stream a stream of a particular flow rate flows through two parallel tubes meaning one stream divided to flow in parallel through those two tubes that stream flows a lot easier through a single tube with a with a flow volume equal to the sum of the previous two um in the um second example um if the flow is external to the uh to the solid walls that is for example in two barges that are pulled up The River by um by two TG boats well um the load the total load on those two barges is pulled a lot easier on a single barge that's bigger so in going from left to right in both examples from left to right it is easier to go from two to one and that according to Constructor law direction of evolution from 2 to one um uh suggests that these two examples the uh uh the phenomenon of of accretion or Confluence or Association is uh is part of nature if that is so then why is it that the flows everywhere around us are not of one big size and the reason is that the flows on areas and inside of volume such as the animal body are flows between one point in an area and one point in and a volume in other words flows between one point and very very very many points and this kind of um uh let's call it design challenge is U is satisfied or answered by a flow architecture called hierarchy and uh again empirically hierarchies are everywhere but those who are curious can discover them um one by one and report them in

ways that are memorable and the easiest to me memorize a a punchline is to arrive at the same P punch line many times in presumably disconnected domains so here in in Social organization the uh the um hierarchy of cities in Europe plotted on the on the vertical versus the city rank clearly the big cities are very few the biggest one is only one and then these descending lines best known empirically as zip distributions are um proceeding the same way with a uh with a relatively fixed slope um over time the lines move upward because population has been increasing but the um the hierarchy Remains the Same the zigzag line at the very top is a uh is a prediction of this descending uh Trend one based on a very simple model that I use in my classroom from cities in Europe to trees in the forest um this is a you see this uh uh these this air these two areas that occupy the descending diagonal they represent that based on assuming several patterns of distribution of tree canopies on the forest floor these are three canopies seen from above uh Bas regardless of what pattern of construction you choose the hierarchy is uh is relatively rigid and it looks just like the hierarchy of cities distributed over a continent uh here is another example uh this is in fact a um an icon of the distribution of animal populations on the globe uh population size versus body size it is a descending um Association um which if one gets really curious could punctuate with actual data from the zoology literature clearly we're familiar with aircraft uh airplanes uh are falling on the same line and the P line is that the hierarchy is natural and um and uh the one size is not so uh that's a very important line to keep in mind as we um as we I don't know what I did that I lost my um oh my God I cannot change the slide though right what am I doing uh [Music] um here we go so now I get to the next uh next topic inequality um here's the movement the actual movement on the area um most familiar to us is the movement of freight um it is on an area it has many participants but uh the few the few are large big uh they travel fast and they travel long uh the same Freight is uh is carried by a multitude of smaller Vehicles traveling uh slowly and short distances uh the freight is one the freight is the one that scans the area and uh this coexistence of of the few with the many is in a is stable it is in a stasis stasis it is harmonious um in the animal realm is the same story you see down down the slide the the coexistence of the few large and long and fast with the with the prey that move short slow and the prey is very numerous uh which is a um coexistence that does not change over time uh now in the human realm we Also let's say in our in our lifetime uh the hierarchy of movement on the globe it is in this case represented by Aviation uh the color red is about high density U movement of people people were flying carpets called airplanes and um the rest of the globe is All In Motion it is interconnected but sooner or later uh individuals from the Blue Area have

access to other points in the blue area by passing through the two chambers of the heart of world Aviation so uh incidentally hierarchy in movement is synonymous with hierarchy in the consumption of fuel and what you see on the image has emerged naturally it is not the result a dictate from say the United Nations uh now on the left side we learned that the hierarchy in movement and fuel use is synonymous with the hierarchy in the distribution of Freedom not of of wealth not Freedom wealth on the left side data from from economics uh show the coincidence that um if you plot the wealth on the vertical and the consumption of fuel on the horizontal the um populations of the world fall on the diagonal so uh I regard this um uh first graph on the left side as a an entry in a dictionary which explains that what is known as wealth distribution in economics is known as the consumption of of fuel or for that matter the the distribution of of human movement in engineering and in physics and um social dynamics on the right hand side the same data are plotted with the fuel use on the on the vertical and economic freedom on the horizontal and uh the arrow pointing toward the future because by the way all these dots are racing upward meaning listen to mental reports they all are about uh this year being better than last year I mean if if the people get lucky then on the right hand side you see that the dots are the dots are M migrating toward um toward the greater Freedom uh this is obvious graphically but no not known very well um uh to the to the Bro to the to the general public now uh looking more closely at what's going on uh in every such country Advanced or not so Advanced the population uh has uh unequal uh access to to wealth and um uh this is a common uh presentation of the distribution of wealth in society on the in on the vertical direction is the the um um accumulated share of income that is a share that that belongs to a partic fraction of the population so the income share is called Epsilon the fraction of the population is beta we'll see this graph more than once after this and if the distribution of income were to be uniform over the population then the curve should be the straight diagonal but none of the curves are straight they are in fact convex with say France and Germany being relatively close to the diagonal uh the US and China uh although similar to each other they're uh more convex and uh brazil and South Africa arguably less developed than the previous four um have curves that are even more convex meaning devia deviating from the diagonal um we're going to see very soon why uh this uh convex curve is natural um and um I decided to illustrate or to present the answer by imagining a population unlike in this graph about the four six countries a population of things that are not do not represent uh human society and um I have on the left side images of the so-called river basins beginning with the uh actual river River delta then the River Basin of air flow through the lungs uh River Basin of water uh

being pumped up upward by the the tree in in Africa and then in the bottom photograph the River Basin of skars coming down the slope toward the Chalet um they're all hierarchical and their convex curves convex curves are crowded in the lower left corner of the Epsilon versus beta graph in other words by the way the the rest of the picture shows details of that crowdedness for the details you have to flip through the book the point is that in the absence of people in the population on this flowing system the um the inequality is U is uh much much much worse so uh uh social existence meaning uh living together has brought these curves upward from the lower right corner uh in the direction of the diagonal this uh this is in my opinion good news but it is relatively unknown In the comparison between the living society and the by the way they're all life FL systems but the comparison between a human society and a society of things that flow without people connected to their neighbors is that difference is the Striking this brings me to two examples of social organization and how it benefits from in Innovation the first example is a look at the prehistory from left to right on the in the vertical direction that the Q represents in dimensional terms represents the the rate of U of um um loss of heating from a uh container with hot water hot water being used by primitive uh population on the abissa that's the horizontal direction is that m TAA is dimensionless and is proportional to the rate of hot water consumption by indivi by one individual it means that on the Absa the direction of time from prehistory to modern times is from left to right now if you look at the piecewise linear curve that emerges it starts on the left with um with the the um hunting and Gathering individuals living in in Huts one pot of hot water meaning on fire for one particular Hut and that pot leaks heat to the ambient and that rate of heat leak is proportional to the Q on the ordinate um now the invention the first one is uh where was the idea of having a larger pot of the hot water um as shown in the example called n equals 3 and then from that pot the individuals are drawing their own supply of of hot water and then climbing up the curve one invention after the next leads to social organization or leads to Association more more individuals or more neighbors uh come com together and benefiting from one Central uh uh source of hot water and what's interesting here is that the um from one invention to the next the slope of the curve uh decreases and that means that on a per individual basis it becomes less uh uh uh costly or uh the loss diminishes and therefore everything about hot water becomes more economical the next graph is about another invention called the dendritic design as opposed to radial design so the dendritic design um is a an invention that allows the user to to get by with by paying even less for uh heat loss to the ambient in other words the the making of hot water B becomes more economical the radial

designs envelope is at the top and you see these thin lines uh they have to do with the invention um and the the thin lines are uh descending they are sliding downward as the size of the population called the capital L increases a second example of uh uh what an invention does uh for the benefit of the whole population is here um I proposed this uh uh image of of complete complete equality um of a population um it's spelled out at the top you have a square domain populated with equal inhabitants equal inputs of something for example water or hot water or whatever uh milk if you want uh and equal channels um and equal channels are bathed by equal flow rates and this is this area is a um an area that facilitates the flow of this something from the area to the point and the point call The Source or the the sink is in the lower right corner um and going from the upper right to the lower left the um the grid becomes finer which means that the population increases on the Square and um in that direction the um the convex curve of Epsilon versus beta migrates toward the corner in the lower right in other words the distribution of this flow over the pop population becomes less and less equal um and that direction is the direction of the increase in the popula the size of the population um so this is the first conclusion uh next is what can be done about it um and by the way U you may have thought of Communism if you look at the first paragraph um the this imposed equality on on the population is uh is easy to imagine maybe kind of more difficult to implement but that has been tried but when there's movement from the area to one point sorcery sync uh clearly clearly the individual um in the vicinity of that very special Point benefits from a bigger beta a bigger beta U corresponding to anyway very few people like himself or herself anyway what can be done about it so assume that the design has only two * two participants as in the upper right that simplest square and that simple square is investigated here first um it is reproduced in the lower right corner and it's labeled a design without innovation um to to the left there are two designs in which one Innovation was implemented and that Innovation is represented by the solid Arrow the solid arrow is the idea of somebody to reduce the flow resistance between two neighbors so um in other words the flow was liberated by um by one inventor uh with regard to his or her access to uh an adjacent neighbor and you see here that um uh regardless of where this inventor was located on the Square the effect of liberating the flow is to move the curve that curve to move it closer to the to the diagonal so uh a single Innovation is a local design change that uh benefits uh the entire population not only the inventor um next uh again all these examples have to do with changes changes that happen in in uh flow rates and the distribution of flow rates called movement and wealth and all these other things including Freedom over the population another way to characterize all these things the hierarchy and the

inequality and so on uh is the word diversity and here I um show several images of diversity in what concerns us the most which is Human Society uh this is my um um graphic way of putting in front of you where the homo sapiens came from um this is a picture of the past two or three millennia um it began with Theus and then the spread the first migration was uh 2 million years ago uh the next uh the ,000 den was 500,000 years ago um and then the rest you can leave the labels on the uh on this uh on this uh Fork shaped flow architecture what I think you should pay attention to is the the word artifact which is in the red at the very top all this movement uh movement toward the uh div diversification of the of the population of homo sapiens had to do with with change change which means if you want inventions uh on the on the obsa on the timeline fire fire was an add-on that um empowered the prehistoric human next and the fire when because of fire the first migration Out of Africa became possible next came clothing clothing made the second migration possible clearly in the direction of of North and Northeast uh Europe and um Eurasia Siberia uh clothing big deal uh fire and clothing both have to do with power power and the heating or the retention of of that heating and then the third um power related change or artifact was boating boating which enabled uh these early humans and hominids to uh to invade the the world of the world as we know it the result as you see the horizontal line is the diverse population that characterizes uh the globe today and now all that means that the mod modern humans are the object of the history that is taught in schools that history is only the blink of one eye 10,000 years um no more and in this direction more of the same took place you look at these artifacts on the timeline at the very bottom agriculture the wheel domesticated animals and then of course the uh the mechanism the machines language of course and literature uh printing press and all today all the way to today's uh advancements in human affluence which are in fact possible because of the Industrial Revolution um and now U coming even closer to who we are and closer to the present science itself is an evolutionary design that um that um took place and continues to take place because of freedom freedom to uh inov innovate freedom to communicate and freedom to implement my own special is thermodynamics um this is relatively new science 50 years since it it started to spread seriously before theonomics the science had very strong legs uh all of them successful uh the heating part was called caloric Theory uh the rest of it without heating was called mechanics this is as old as as even before the Greeks the old Greeks and of course uh Evolution which really means design is also even older than the previous too we see it in in everything from cave paintings to to the Renaissance and of course uh uh the kind of U you know uh designing and the construction that's going on today along the way the science

the science became unified uh with a construct the law uh the branch called the heat transfer where um thermal engineering joined the rest of the uh the uh the theory and became uh what I'm what I'm teaching today and what I think is very very powerful science of thermodynamics with uh design evolution in it so um where is this uh uh arrow pointing obviously it's pointing toward the future the um but the present unfortunately is a little bit complicated um let me go back to the previous story if you read any book of the history of physics or the history of thermodynamics you'll find um a story about individuals the uh the very few the very few beginning with Sadi Carnot and his father was called named Lazare Carnot um before that of course Archimedes and many many of these very very few uh distinguishable names um and um design Evolution has its own luminaries individuals individuals who um lived and uh created in places that were known known for uh creativity and for the uh for the sanctity of the individual uh places geographically but also places in the history of the human civilization uh the names of these individuals uh speak of uh who they were where they lived and what language they spoke well um less and less of that is uh is uh is the character of what goes on today in science this is a cartoon from the very popular French magazine called *Le Point* uh which is something like between American scientist and the uh Scientific American you see here um yeah a joke about uh what you what you see published these days articles coauthored by uh 10 or 20 or sometimes 40 co-authors 40 co-authors um which give the impression or attempt to perpetuate the uh the The View that better science comes from the larger group uh that is a view that's not original uh it happened with the collectivization of Agriculture and um the history of the 20th century is very clear about the uh the uh the the success of that success in quotation marks well um uh this is the present and luckily in it uh this is what's going on in it there are individuals just like in the books of uh of the history of of science uh who by their very nature um do not feel at home um uh Marching In columns they simply run away if they can survive or they shy away uh they the they are not joiners they are happy with their ideas because uh only that way uh they can uh climb peaks unlike the margin columns so I end with uh the view of this little man conquering Mount Everest I drew I made a drawing in 1975 with a big pen in one stroke um 1975 is when I was uh uh leaving uh MIT to become a post doc in Berkeley uh and I uh I that that uh caricature is Adrian um attracted by by the idea somewhere at the very top and this is on the left uh uh how it how it looks from that very high position uh the single investigator the one with the driven by ideas uh does two things and both are uh comforting one is uh he steps on Deadwood Deadwood that's a metaphor for everything from old ideas and inactive or groups that are not creative but in front of the

individual there is a an infinite uh Blue Sky of opportunity in which the ideas could be projected tried and implemented so this is the thread of the book once again on the right hand side and thank you for your attention thank you thank you Adrian should I stop sharing um you can leave it okay Susan would you like to begin sure yes sure um so what I'd like to do is is kind of share some insights that um that have occurred to me me and you get you Daniel and you Adrien um of whether you know I'm I'm on the evolutionary track right or you or it'll give us something to riff about um um so as I reread this and I have a an autograph copy here um it really the impact became much more powerful for me because one I started understanding that there's a correlation now there's a difference in the language here but there's a correlation Daniel um that um active what Constructor law does is actually validate um active inference because what Constructor law is saying is that you nature is going to evolve um based on minimizing heat transfer or heat loss and free energy is saying the same thing with you know we are driven to um minimize our free energy oh no no no no okay so so so hold on a yeah said that yeah I know that these are two different terms but I'm I'm just trying to connect this for the audience um and um and so I have some really specific questions for you so I want to I want to kind of stay stay on back here not get caught up in in too much of the definitions because one is the audience that's going to watch this they're probably not going to be as um astute about the the hard science of thermodynamics um versus you know the uh the machine learning the um the biology the psychology the philosophy that you know that we really talk about here so I'm just kind of threading that need need um just so when I when I kind of start C start trying to capture the essence of the beauty quite frankly of the Constructor law they're able to connect those those blankets okay um okay let me let me let me respond to what you bring up Okay I uh I think more highly of the audience I think the audience is stood and even even if even those like Adrian who are not experts in a particular uh corner of science are able to connect the dots and this is why um U my way of teaching is to use um common language whenever uh whenever possible uh and that is the direction in which I I want to to push our dialogue here uh I want to I want to respond to a few things that you said first uh design in nature evolution is not about minimizing heat loss it is about providing uh meaning evolutionary design changing design morphing design uh to u to facilitate uh flow or movement and in every uh example of mine um the uh the Mover the Mover the answer to the question what moves was the IND idual or the group any yes the individual which is why I started with hunting and Gathering benefited as a mover meaning as a living uh body benefited from uh not wasting U uh heat from the pot of hot water okay

meaning the individual benefited from being economical here you see the word being economical or being Thrifty as another way of expressing the natural tendency uh encapsulated in the Constructor law so that that is that is that is the that is design in nature second design in nature has nothing to do with what you call free energy it is about it is about this moving architecture that keeps on changing and yes uh in order to move the object I call it theonomic system the The Entity must be pushed and the pushing comes from power and the power comes from uh architectures most of them natural some a few of them made by people called engines were functioning as engines what those systems consume uh from animals to uh to uh to people and to the entire Globe um are called food fuel and solar U solar input um that is uh and and those things those things are not free those things are not free uh I can give you examples of how dear every single one is uh and finally finally um you and I had a conversation via email if the audience would like to find out what free energy means I would be happy to come back and give you uh again common common sense uh or uh common language uh presentation of what is meant by free energy in thermodynamics um it is a pretty good name for something that was coin around 1860 1860 but no free energy is not free okay and and it is not responsible for setting things in motion or for creating change changes quite the opposite free energy is a is a um just like entropy mathematical formula uh useful in the description of a of a of an entity that is in perfect equilibrium with its environment in other words an entity that is not pushed by the surroundings or pushing the surroundings uh but it is useful for for other reasons which require a hourong discussion um and yes some illustration that I offer to to produce if if people are curious uh but by the way U I'm not criticizing anybody and certainly uh free energy principle is something else not free energy okay so that's the uh keep in mind these are two different Notions and uh but it's very important to to to be very clear about one about the the fact that in our conversation we uh we respect the meaning of the words absolutely and thank you for that because I mean that actually um Rich enriched my uh the how I'm tying or banding these together I kind of like to point to the economical because that actually makes a lot of sense but in reading the book what I did get a distinction of and again you I want to hear correct me that's not a problem is that you know when we're thinking about active inference and you you know we're really more focused on you the agents and the and interactions with the agents and what I what I um what occurred to me especially as I was looking through your slides Adrian is you know is that the flow architecture um one it's it's kind of pointing us toward what is actually flowing and so many of our mental models are static so this is where I think that it you do have to really wrestle with this but

understanding how nature is um is designing and cesing um and how we as interactive agents in these flow architectures are um are reacting to the um the the natural things I mean in other words in in in condensing the information I was like okay the flow architecture is the natural things that are observable palatable movable and measurable okay um and um and the the flow Direction and the configuration and the time I.E rhythm is actually what is what needs to be free um and and I pulled one thing that I thought was really rich in terms of um these features which again are contextually bound but they're flowing together and one point I wanted to make was is that they have to be or or ask you about in terms of how are those features needing to be independent first and number two is um is how they are flowing together you pointed at it in one of the slides here but the how they're actually more economical when they are powering together well I think is a you a really key key aspect here let me let me respond it is uh it I confess early on that's 25 years ago um coming from engineering i u i saw I saw flow first before I saw the configuration or the design flowing or changing or morphing the phenomenon is uh is not the the flow and the direction or for that matter the uh the decrease in the so-called flow resistance the phenomenon is the um the um um tendency of the um of the design to change all of it to change so that so that that movement which is belongs to the body but also to the niche around it that movement is facilitated um a friend of mine the other day brought up the example of the U one of the valves moving and flapping inside the human heart uh yes that is the valve is the niche the flow is the blood flow the blood flow has edies the blood flow is constantly morphing keep in mind the the flow of liquid is uh is uh is complicated it's like stuffing a rope down the throat of a gun barrel it's a it's it has configuration that's invisible to to most Mortals and in a case of the uh of of the living organ the heart uh the stuffing of the Rope called blood flow uh brings with it the uh the rhythmic the loving uh deformation of the of the flap of the valve and U and this is all for the benefit of the moving animal yes the freedom and the the life I call it access according to construct the law uh the freedom is freedom Galore in the description of what I just told you there are innumerable degrees of freedom you know the bending of the of the flap the direction of the flap the rhythm of the flap and that's only about the flap but when I get into the edes inside the blood flow well those are innumerable once again hierarchical a few large and many small and they move these blobs moving speak of Rhythm but if you take a snapshot you see hierarchy uh yes it's complicated but it happened naturally uh and the direction of the happening is toward making a better and better animal a longer living animal a longer traveling animal that is the uh the and and why should the animal evolve in that direction because the animal

along with with with the with his or her compatriots but is a participant in the River Basin of water called Humanity humanity is spinning water on the globe and morphing and evolving and getting okay better at moving itself which means moving water just like any River Basin or just like the hydrosphere and uh it's all a relatively boring conclusion because it's always the same conclusion but it is a beautiful conclusion because it makes us on the one hand not waste time on something that repeats itself and gives us the opportunity to come up with with better ideas or new ideas or frankly to make D big bigger noise to say aha Aha and I'll say another thing before I stop um you brought up more freedom yes it's very important in um in my okay modest way of teaching these things I show what happens if you have a one of these morphing designs uh U any anything such as a very simple drawing of of a River Basin and what happens if you uh uh increase the number of degrees of freedom from one one version design to the next so I show this is with pencil and paper that as the number of degrees of freedom increases the access which is measured in many ways um access increases if What flows is people walking through the streets and everything including vehicles that means wealth increases uh in the direction of Greater freedom and this is why it's no it's no coincidence no coincidence that that the distribution of innovation on the globe is uh is nonuniform nonuniform just as non-uniform in fact it co coincides with the nonuniform distribution of wealth wealth and the consumption of fuel and all these other things that are that are undeniable so this is the coincidence to keep in mind and keep it in mind when you hear people calling for uh uniform distribution of this and uniform distribution of that and for certain groups to cut down in the consumption of fuel and to stop using you know commercial Aviation because they are burning too much fuel yeah think about that think about how intelligent that the proposal would be and by the way uh the reason is not intelligent is because even if if you implement it it is calling to be replaced tomorrow because that kind of change is unnatural you see I'm not okay so um so here's where I'm confused um and so help me help me to kind of pull this together because we talk a lot about um about self organizing systems and and about degrees of freedom and what I'm kind of teasing out and you maybe I'm confused maybe not but you what I'm as I'm digging into this and really you appreciating the richness of of your Precision of your language um I want to I want to kind of capitalize on um on as as the evolution occurs um and um and how going back to the inequality and how the system is natural natur AI going to lead to the different the different areas and volumes that the point of the area and the volume I think is I think that's one of the richest things that probably the hardest to grasp in here but but yeah but the also the more volume there is and this is a

question um then the more um coalescence the um uh accretion I believe is increases but that also leads to fragmentation is that how does that relate to the the point in the area let me let me address that you this big volume you referred to um is one um one feature of the object or of the design is called size size everything that we uh use in our discussion today is about the a thing with many features including size size is a natural feature uh I I started that way not not not everything is big or not everything is should be this small yeah everything should be this size and if you if you look carefully at the at the literature That Grew around the construct law you'll see many many predictions of what the size of the bird should be the size of this airplane should be and the size of this helicopter should be because size is a a natural feature of the moving flow system and U so uh what moves is uh is a more complicated story than accretion is comes from celestial bodies where where let's say particles in the primordial soup attract each other and they come together or forget about the uh the uh the thing that nobody saw nobody saw the primordial soup but you can look at you can no you can actually look at uh at the little uh suspensions of dust and how they uh they uh become Aggregates dust Aggregates shaped as just like the snowflakes snowflakes uh you see this in Hidden Corners somewhere in a room where no one has been for about 30 years uh these dendrites of of particles that came together because of mutual electrostatic attraction and um yes but well those systems by this by this time are dead they are not the object of our our discussion although on root to those D rights they were alive and behaving just like any other any other flow architecture in their case proceeding from a volume to one point um so I come to the um uh anyway the observation is that the the design in nature is uh is is not about uh accretion that's just one one kind of flow uh it is about uh the the whole architecture that has this purpose of enabling uh easier movement for the whole the whole I think okay what's difficult here I I recognize where my peers feel difficult to grasp the difficulty is is uh to to get away from the um from the the present tendency of of um splitting hairs called U you know uh respecting too much the infinite decimal it's called reductionism uh this discussion about design in nature is about the whole it's about the whole uh is about the rising above uh above the ground and and um and engage in holism holism and U or getting away from disciplines and becoming a polymath okay uh it's about and if you if you become a polymath you become closer to being a street person a person who looks at this poster this poster reads 10 different books 10 different books not just narrow narrow narrow narrow and a person who does not live in a a in a cubicle or in a locker room a person who has many friends many friends a person who knows more than one language a person yes U this is the the meaning a truly truly

common person and um that's the invitation to the scientists is to to to to get out get out of the department you know physically or figuratively and smell the roses uh because beauty beauty is uh where you don't don't expect to find it okay meaning it's a it's maybe maybe behind the house of the neighbor that's probably where the beauty is so this one because this one's so rich but I could go on for hours I'm gonna turn this back over to Daniel yeah because thank you so much because this was been thank you Susan just in our last minutes um I'll just add a few small points um first just beginning with the last common person is that what is held in Commons a Commons person and the stewardship of the commons then in your drawing with the mountain there was the peak up and there was the path out and it was like a person walking on a path forward out in a river flow and time as you began with is goal flow it flows in the goal oriented Direction yes there's more to it and it's flowing balls roll downhill and uh the the organism walks forward it can't walk backwards uh and then the last point is the uh distributions empirical that you presented um Jeffrey West and others have highlighted that empirical aspect of the outcomes and their uh distributions however you tied that to the 0 to 1 to two to three dimensional um transfer points where from one to two is like a point spreading out on a plate and threedimensional is like a point radiating out these are fundamental equations that are kind of in your engineering tool kit because they have to do with like for example heat distribution but that's just related to the modes of vibration and the degrees of freedom of the molecules as a secondary distributional approximation to an actual size specific and that's why there are river bifurcations that are this big and not river bifurcations that are a micron and all these other kinds of things that are um not just the neutral we need to explain away about nature but just straight down the middle with what the study of nature is that's right uh yes I couldn't agree more and I get the I Got U uh enlightened by by this kind of commentary um and um and yes you mentioned Jeffrey West he is one of the few uh uh in whose company I feel at home uh they they U it's important it's important to uh to uh once again to look at the uh at life from both sides okay is in the song uh and um and uh and make a very strong effort not to become an ideologue uh um I um I like the opposite of that which is that uh that uh my students idea may be better you see where maybe uh I get corrected every I teach this course right now I teach it to a an overflow crowd um and quite quite frequently uh not every day they correct me they say how about this and it's uh it's it's like a free food for me the sort of reaction uh uh the listeners should pay attention I don't get these comments from my colleagues okay I get them from these nobody's from these nobody's called students yes don't say that too loud it's a perfect it's a perfect note and it's one of the

things I've always loved the most is outside of the organizational structure and the shape of the building is about the fact that it's first principles and about the opportunity though unlikelyhood or not of a transformative epistemic relationship across all language and all professional boundaries so thank you Adrian for for also living this and sharing how you've done it yeah yeah thank you so much this has been awesome thank you to both of you till next time bye bye